Machine Learning Methods for Solving Fire Detection Problems

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Received 21.06.2025, received in revised form 08.11.2025, accepted 15.01.2026

Abstract. This article presents an analysis of the machine learning algorithms «Random Forest», «Logistic Regression» and «Convolutional Neural Network». A review of scientific and technical literature on the topic of the study is conducted, the key advantages and limitations of the considered algorithms are analyzed. Particular attention is paid to the application of algorithms in image classification problems, data collection and preparation of satellite images for training. The article contains a practical implementation and comparative analysis of the efficiency of the algorithms in various conditions. The purpose of the article is to provide an understanding of the mathematical foundations of the algorithms and the practical aspects of applying machine learning algorithms to detect fires in satellite images.

Keywords: logistic regression, random forest, convolutional neural network, satellite images, wildfire, spectral channels, error matrix, ROC-AUC metric.